

Anubhav Jangra

Computer Science Ph.D. student, Columbia University

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Education

Present Aug 2023	Columbia University Doctor of Philosophy in Computer Science (Advisor: Smaranda Muresan)	GPA: 3.86/4.0
Jun 2021 Jul 2017	Indian Institute of Technology, Patna Bachelor of Technology in Computer Science and Engineering	GPA: 8.82/10

Research Experience

Aug 2024 May 2024	Microsoft Research & Office of Applied Research <i>JEM (Joint E+D & MSR) Research Intern Supervisors: Sujay Jauhar, Adrian de Wynter, Bahar Sarrafzadeh</i> Incorporated style personalized text generation (SPTG) into Word CoPilot's draft and rewrite text generation features to reflect users' authentic voices. Successfully launched the prototype for internal dogfooding. Investigated automatic evaluation metrics for low-resource SPTG across multiple writing domains and evaluation settings [under submission at ARR].	Redmond, USA
Jul 2023 Jul 2021	Google Research Advertising Sciences Team [🌐] <i>Pre-Doctoral Researcher Supervisor: Aravindan Raghuv eer</i> Explored NLG techniques for creative advertisement generation. Investigating several research areas like text style transfer, data-to-text generation, automatic code generation, semantic representations etc.	Bangalore, India

Publications

US=under submission, P=Preprints, C=Conference, B=Book, SP=Short Paper, J=Journal

Selected Works

- [US] **Evaluating Style-Personalized Text Generation: Challenges and Directions** [🌐]
[Anubhav Jangra](#), Bahareh Sarrafzadeh, Silviu Cucerzan, Adrian de Wynter*, Sujay Kumar Jauhar*
ACL Rolling Reviews October Cycle [ARR'25]
- [US] **Designing and Evaluating Hint Generation Systems for Science Education** [🌐]
[Anubhav Jangra](#), Smaranda Muresan
ACM Conference on Human Factors in Computing Systems 2026, Barcelona, Spain [CHI'26]
- [P] **Navigating the Landscape of Hint Generation Research: From the Past to the Future** [🌐]
[Anubhav Jangra](#), Jamshid Mozafari, Adam Jatowt, Smaranda Muresan
Transactions of the Association for Computational Linguistics (presented at ACL 2025, Vienna, Austria) [TACL'25]
- [C] **Large Scale Multi-modal Multi-lingual Summarization Dataset** [🌐]
Yash Verma*, [Anubhav Jangra](#)*, Raghvendra Verma, Sriparna Saha
The 17th Conference of the European Chapter of the Association for Computational Linguistics, Dubrovnik, Croatia [EACL'23]
- [C] **T-STAR: Truthful Style Transfer using AMR Graph as Intermediate Representation** [🌐]
[Anubhav Jangra](#)*, Preksha Nema*, Aravindan Raghuv eer
The 2022 Conference on Empirical Methods in Natural Language Processing, Abu Dhabi, UAE [EMNLP'22]
- [J] **A Survey on Multi-modal Summarization** [🌐]
[Anubhav Jangra](#), Sourajit Mukherjee, Adam Jatowt, Sriparna Saha, Mohammed Hasanuzzaman,
ACM Computing Surveys [ACM CSUR'23]
- [C] **WIDAR - Weighted Input Document Augmented ROUGE** [🌐]
Raghav Jain*, Vaibhav Mavi*, [Anubhav Jangra](#)*, Sriparna Saha
44th European Conference on Information Retrieval, Stavanger, Norway [ECIR'22]
- [C] **Multi-modal Supplementary Complementary Summarization using Multi-Objective Optimization** [🌐]
[Anubhav Jangra](#), Sriparna Saha, Adam Jatowt, Mohammed Hasanuzzaman
44th International ACM SIGIR Conference on Research and Development in Information Retrieval, Virtual [SIGIR'21]
- [C] **Semantic Extractor Paraphraser based Abstractive Summarization** [🌐]
[Anubhav Jangra](#)*, Raghav Jain*, Vaibhav Mavi*, Sriparna Saha, Pushpak Bhattacharyya,
17th International Conference on Natural Language Processing, Patna, India [ICON'20]
- [SP] **Multi-Modal Summary Generation using Multi-objective Optimization** [🌐]
[Anubhav Jangra](#), Sriparna Saha, Adam Jatowt, Mohammed Hasanuzzman,
43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, Xi'an, China [SIGIR'20]

- [SP] **Text-Image-Video Summary Generation using Joint Integer Linear Programming** [🔗]
 Anubhav Jangra, Adam Jatowt, Mohammed Hasanuzzman, Sriparna Saha,
42nd European Conference on Information Retrieval, Lisbon, Portugal [ECIR'20]

Other Works

- [B] **Multi-hop Question Answering** [🔗]
 Vaibhav Mavi, Anubhav Jangra, Adam Jatowt
Foundations and Trends® in Information Retrieval Vol. 17 Issue 5 [FnTs, 2024]
- [C] **TriviaHG: A Dataset for Automatic Hint Generation for Factoid Questions**
 Jamshid Mozafari, Anubhav Jangra, Adam Jatowt
The 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, Wash. D.C., USA [SIGIR'24]
- [C] **Can Multimodal Pointer Generator Transformers produce topically relevant summaries?** [🔗]
 Sourajit Mukherjee, Anubhav Jangra, Sriparna Saha, Adam Jatowt,
2023 International Joint Conference on Neural Networks (IJCNN) [IJCNN'23]
- [C] **Topic-aware Multimodal Summarization** [🔗]
 Sourajit Mukherjee, Anubhav Jangra, Sriparna Saha, Adam Jatowt,
2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics [Findings in ACL'22]
- [P] **A Survey on Medical Document Summarization** [🔗]
 Raghav Jain, Anubhav Jangra, Adam Jatowt, Sriparna Saha
ArXiv 2212.01669 [ArXiv, 2022]
- [C] **Combining Vision Language Representations for Patch-based Identification of Lexico-Semantic Relations** [🔗]
 Prince Jha, Gaël Dias, Alexis Lechervy, José G Moreno, Anubhav Jangra, Sebastião Pais, Sriparna Saha
30th ACM International Conference on Multimedia, Lisbon, Portugal [ACM MM'22]
- [C] **MAKED: Multi-lingual Automatic Keyword Extraction Dataset** [🔗]
 Yash Verma, Anubhav Jangra, Sriparna Saha, Adam Jatowt, Dwaipayan Roy
13th Conference on Language Resources and Evaluation [LREC'22]
- [J] **Identifying Complaints based on Semi-Supervised Mincuts** [🔗]
 Apoorva Singh, Sriparna Saha, Mohammed Hasanuzzaman, Anubhav Jangra
Elsevier's Expert Systems with Applications, Volume 186, 2021 [ESWA'21]
- [J] **Extractive Single Document Summarization using Multiobjective Optimization: Exploring Self-organized Differential Evolution, Grey Wolf Optimizer and Water Cycle Algorithm** [🔗]
 Naveen Saini, Sriparna Saha, Anubhav Jangra, Pushpak Bhattacharyya,
Elsevier's Knowledge Based Systems, 2018 [KBS'18]

Other Experiences

Aug 2020 Jul 2020	GREYC Lab, ENSI-CAEN [🔗] Research Intern / Supervisor: <i>Gaël Dias</i> Extended patch-based lexical semantic identification frameworks to a multi-modal setting. Developed the dataset and conducted the pilot studies of the project. [ACM MM'22]	Remote / Caen, France
Jun 2019 May 2019	Graduate School of Informatics, Kyoto University [🔗] Research Intern / Supervisor: <i>Adam Jatowt</i> Explored various unsupervised optimization techniques to develop multi-modal summarization systems that generate text-image-audio-video summaries.	Kyoto, Japan
Jul 2021 Jun 2021	IBM Global Research Mentee / Supervisor: <i>Ganesan Narayanasamy</i> Developed the project framework for <i>Health Care App</i> , that uses knowledge graphs and named-entity recognition to help users self-diagnose themselves.	Remote / Chennai, India
Mar 2021 Dec 2020	Huawei Technologies Co., Ltd Project Member / Supervisor: <i>Sriparna Saha</i> Developed a Proof of Concept (POC) for the task of automatic tagline generation and product description using existing neural summarization systems for the upcoming collaborative project of IIT Patna and Huawei.	Remote

Jan 2020 Dec 2019	TCS Innovation Lab [🌐] Research Intern / Supervisor: Arijit Ukil Investigated generative modeling to tackle the insufficiency of data in time-series signal classification.	Kolkata, India
Jan 2019 Dec 2018	CFILT Lab, IIT Bombay [🌐] Research Intern / Supervisor: Pushpak Bhattacharyya Examined Unsupervised NMT for distant language pairs (Indo-Aryan languages) using attention based seq2seq models.	Mumbai, India
Jul 2021 Jul 2018	AI-NLP-ML Lab, IIT Patna [🌐] Undergraduate Research Scholar / Supervisor: Sriparna Saha Worked extensively in the area of summarization (e.g., multi-modal summarization, extractive and abstractive text summarization), complaint mining and multi-label classification.	Patna, India

Academic Service

PC Member	Coling 2025, LREC-Coling 2024, Text2Story Workshop (ECIR 2023, 2024), IACT (SIGIR 2023)
Reviewer (Conference)	ACL ARR (since Dec 2023), CIKM 2023, ACL 2023, EMNLP 2022
Reviewer (Journals)	ACM Computing Surveys (since Jan 2021), ACM TALLIP (since May 2020), Applied Artificial Intelligence (since Oct 2021), and IEEE Transactions on Computational Social Systems (since Jan 2022), Expert Systems with Applications (since Sept 2022), Engineering Applications of Artificial Intelligence (since Feb 2022), IEEE Internet Computing (since Feb 2022)
Secondary Reviewer	AAAI 2020, EACL 2021, ACL 2021, EMNLP 2021, CIKM 2021, KDD 2022, and WebConf 2022
Research Mentor	Mentoring two masters and two undergraduate student towards building intelligent tutoring systems at Columbia University. Mentored three undergraduate interns, two masters student researchers, and one undergraduate students for automatic summarization at IITP.
Community Service	Creator and organizer of CARE program —a Community for AI Research and Education to guide students and early-stage researchers through their research-related queries.
Volunteer	Volunteered as a reviewer in Google's CS Research Mentorship program to help review applicants from historically marginalized communities for the mentorship program. Reviewer for PhD Pre-Application Review (PAR) program at Columbia university.

Honours and Awards

Google Research AI summer school, 2020 [🌐] One of the 50 participants out of 1000+ applicants in the NLU track.

MSU-IITR-IISc course and workshop [🌐] Attended a short term course and workshop on “Pragmatic Optimization for Practical Problem Solving” conducted by Michigan State university, IIT Roorkee and IISc Bangalore, limited to 40 students.

IIT JEE Ranked in National Top 0.2% (amongst 1,400,000 candidates) in JEE Mains 2017 and Top 1.5% (amongst 2,00,000 candidates) in IIT-JEE Advanced 2017.

Talks

Guest lecture: Hands-on Deep Learning - COMSW4705 (NLP by Daniel Bauer), Columbia University	Oct 2025
Guest lectures: Intro to Deep Learning - COMSBC3705 (NLP by Smaranda Muresan), Columbia University	Sept 2025
Abstract Meaning Representation (AMR) Graphs at work! - AI-NLP-ML Lab, IIT Patna	Oct 2022
Automatic Text Summarization PyData Patna Conference	Dec 2020

Teaching and Leadership Roles

Department of Computer Science at Columbia University Head Teaching Assistant	Sep 2025-present
Teaching Assistant for COMSW4705 (Natural Language Processing) course taught by Dr. Daniel Bauer.	
Barnard College at Columbia University Teaching Assistant	Sep 2024-Dec 2024
Teaching Assistant for BC3997 (Natural Language Processing) course taught by Dr. Smaranda Muresan.	
Google DSC IIT Patna, Patna, India ML Department Lead	2019-2020
Supervised three projects and gave lectures on Machine Learning theory and its applications.	
Univeristy of Innsbruck, Austria Teaching Assistant	Jun 2020
Part-time Teaching Assistant in the course 2021S703836 VU (Natural Language Processing). Prepared lectures on automatic summarization.	